

LIGHTING UP SHMOOCON

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January 10, 2025



TAKE OUT YOUR LIGHT WANDS!



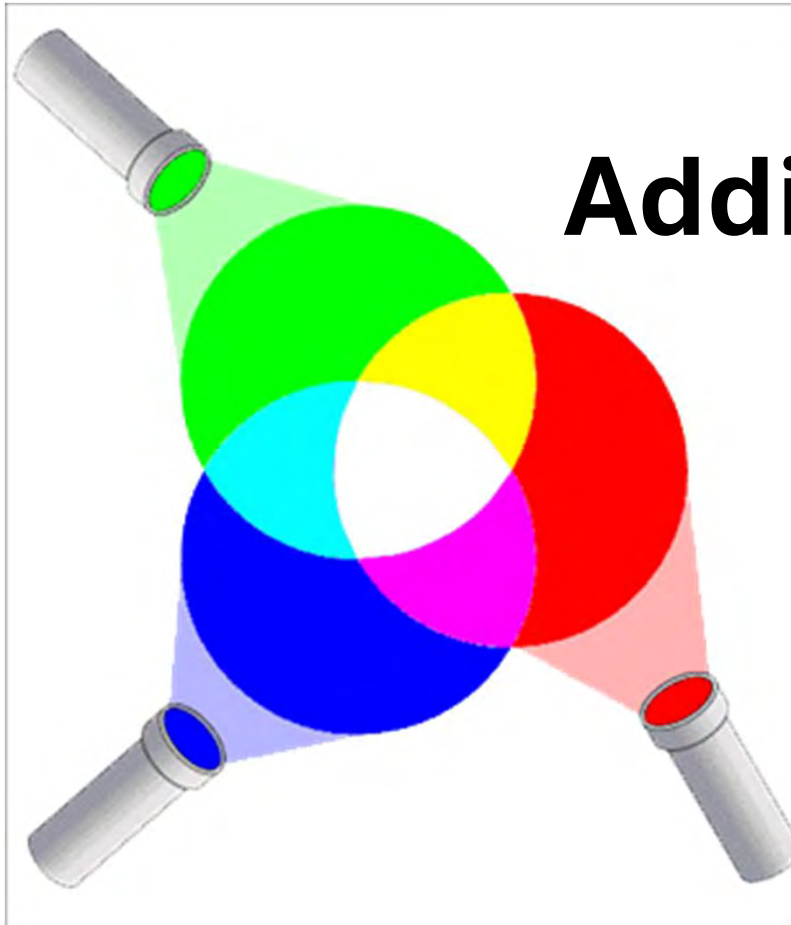
Turn them on with the slide switch (above the round button)



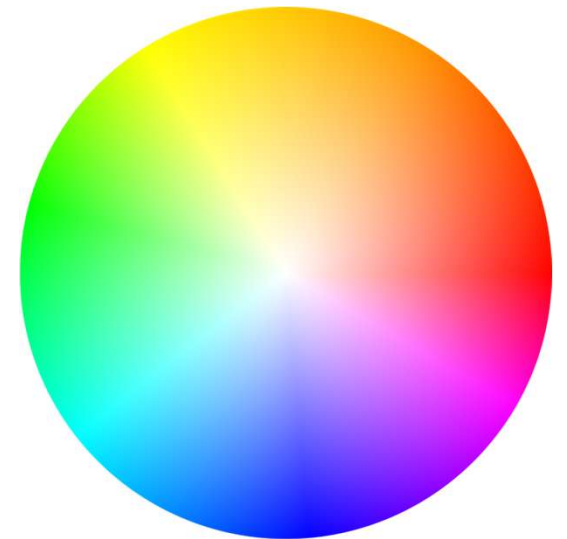
WHAT ARE THEY?

- Five LEDs – operate as one pixel
- 4.5V battery operation (3x AAA)
- Remote control of color
- Microcontroller - Light modes without RF Control

RGB LED: RED GREEN BLUE



Additive Color



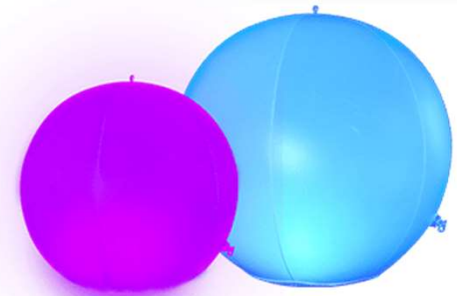


BUILT-IN MODES

- Red
- Blue
- Green
- Violet
- Lime
- Light Blue
- White
- Red Green Blue Flash
- Flowing color change
- Red, Green, Blue Flash 2

FORM FACTORS

- Light Wand
- Bracelet
- Santa Hat
- Drink Glasses
- Balls



DMX CONTROLLER





MANUFACTURER

- LEDGIFTSUPPLIER.COM
- Greatfavonian Electronic CO., LTD.
- Shenzen China



新满杰

GFLAI[®]



GET MORE

- Group Buy / Small Quantities (Mid June)

<https://www.facebook.com/groups/187336816411791>

- Anytime Purchase

<https://wallyslights.com/collections/dmx-products>

- Shmoococon Quantities:

LEDGIFTSUPPLIER.COM

COMPUTER CONTROL

- DMX (digital lighting control standard)





DMX512

- Originally a standardized method for controlling light dimmers
- 512 refers to maximum individual control channels over single cable.
- Expanded with “Universes” of 512 channels

DMX CONTROL

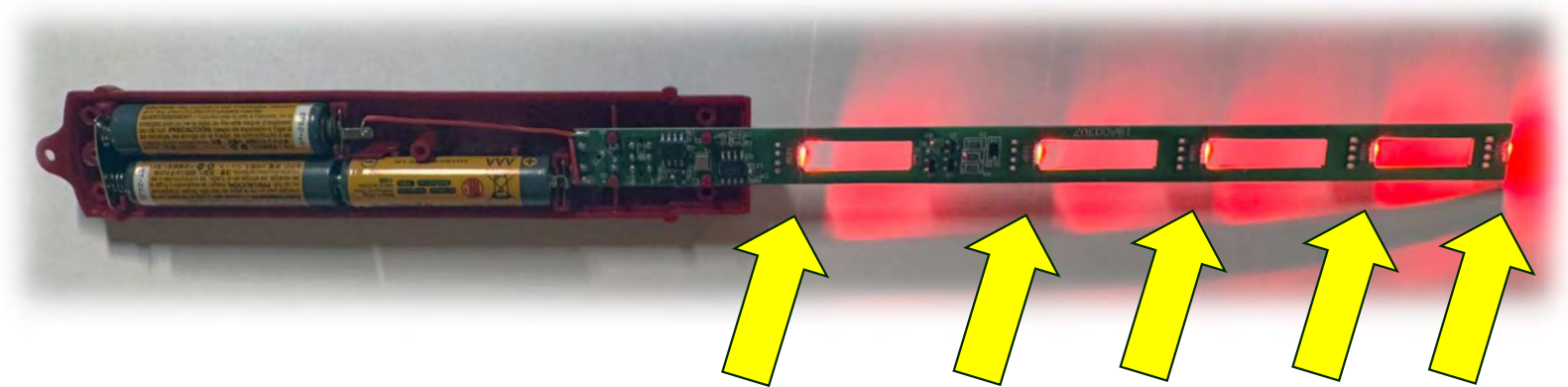


- Decimal 85 / 0x55 ---- Attention, Ch 1
- Decimal 0 / 0x00 ---- Group 0=ALL, Ch 2
- Red Value Byte ---- 0 to 0xFF Intensity, Ch 3
- Green Value Byte ---- 0 to 0xFF Intensity, Ch 4
- Blue Value Byte ---- 0 to 0xFF Intensity, Ch 5

INTERNALS



INTERNALS



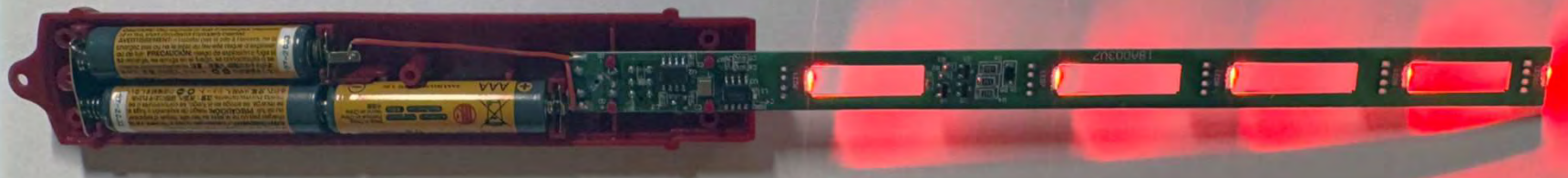
5 LEDs Function as 1 Pixel

INTERNALS



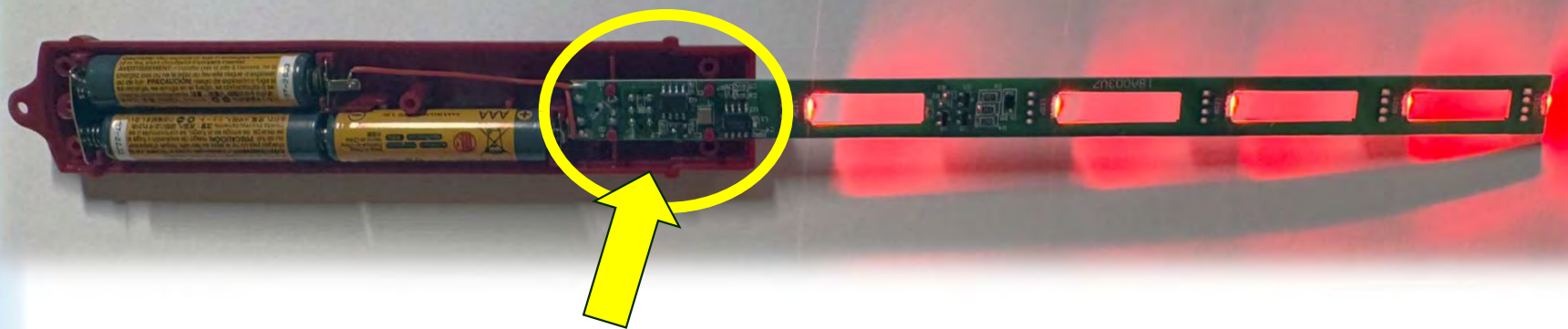
LED Diffuser

INTERNALS



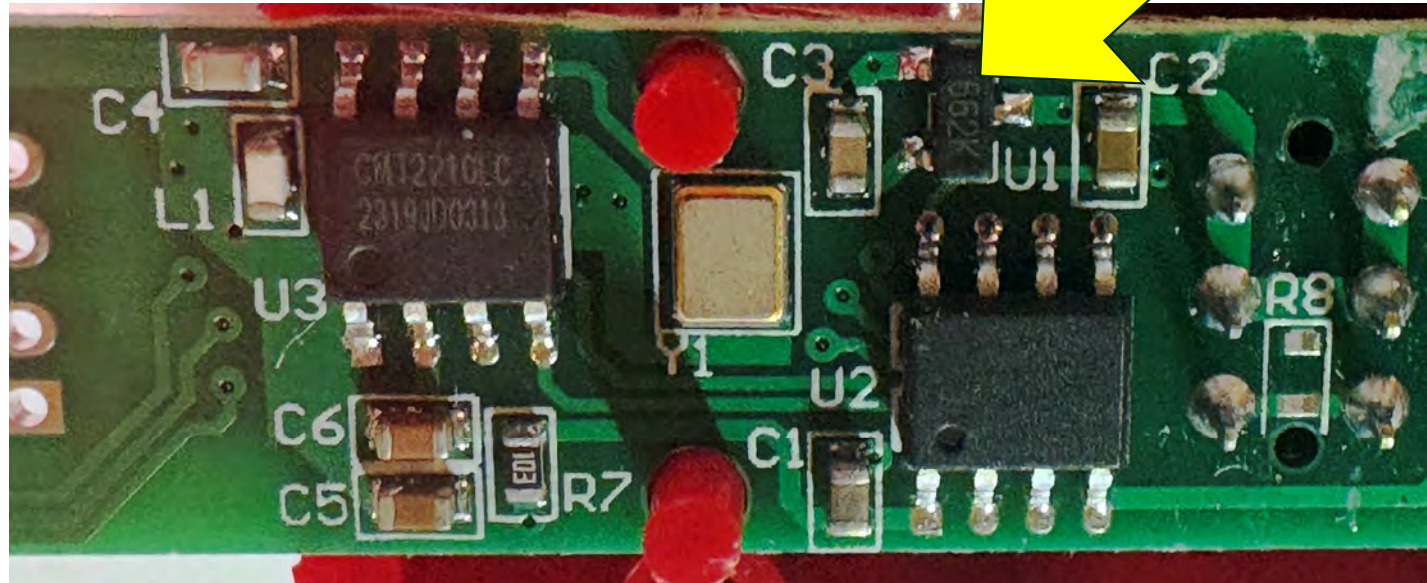
4.5 V Battery

INTERNALS



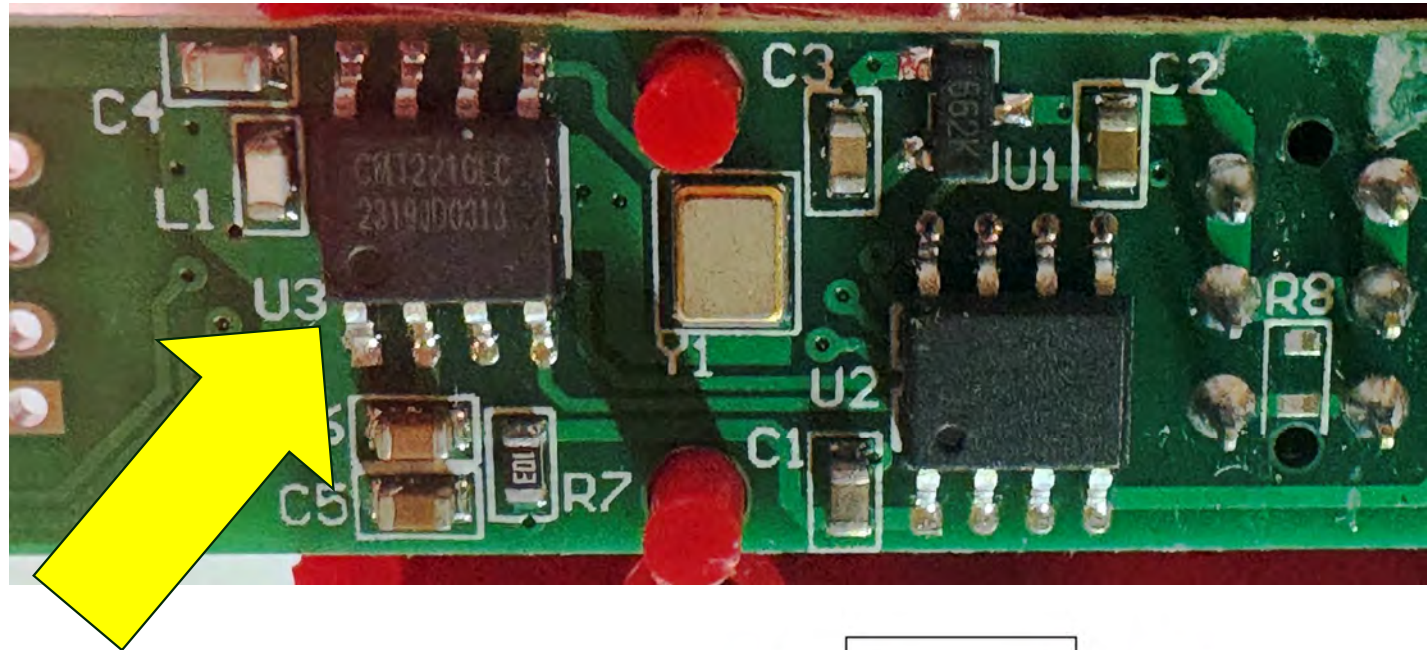
Control Section

INTERNALS



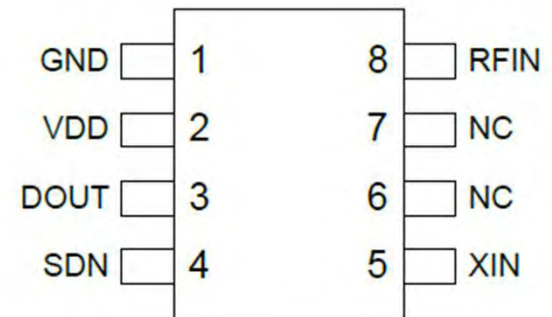
**U1: 3.3V Positive Fixed LDO Voltage Regulator
SOT-23 Package**

INTERNALS

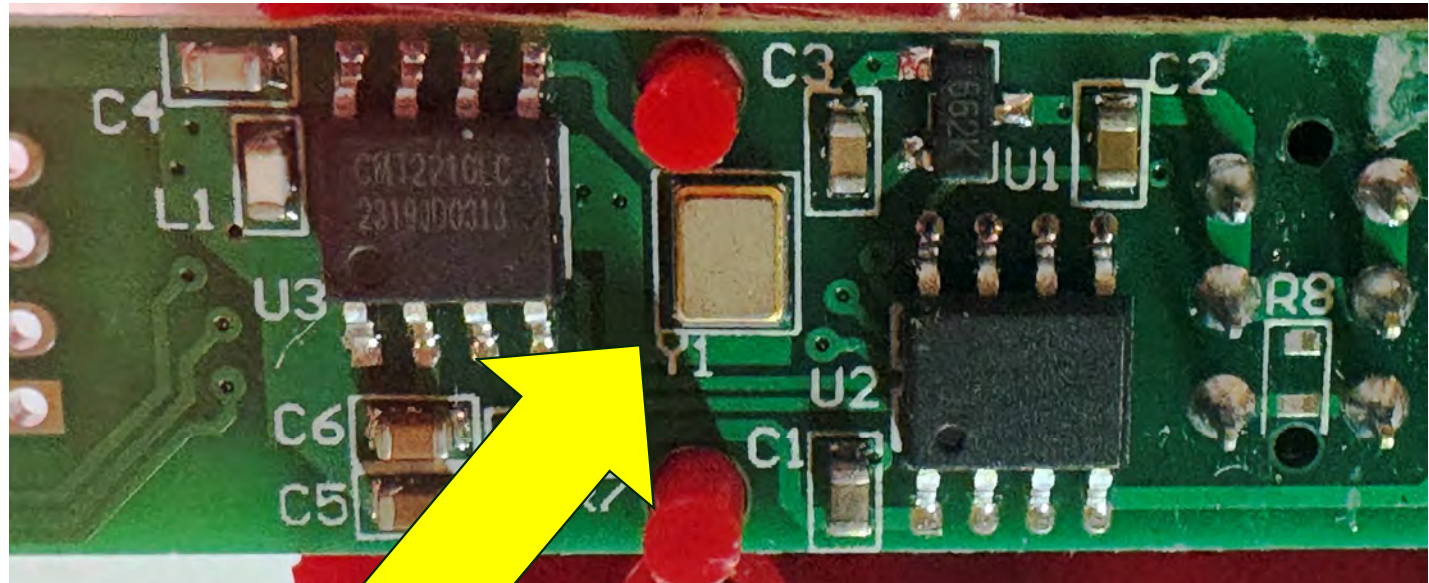


U3: CMT2210LC receiver

**Low power 315/433.92 MHz
OOK Receiver**



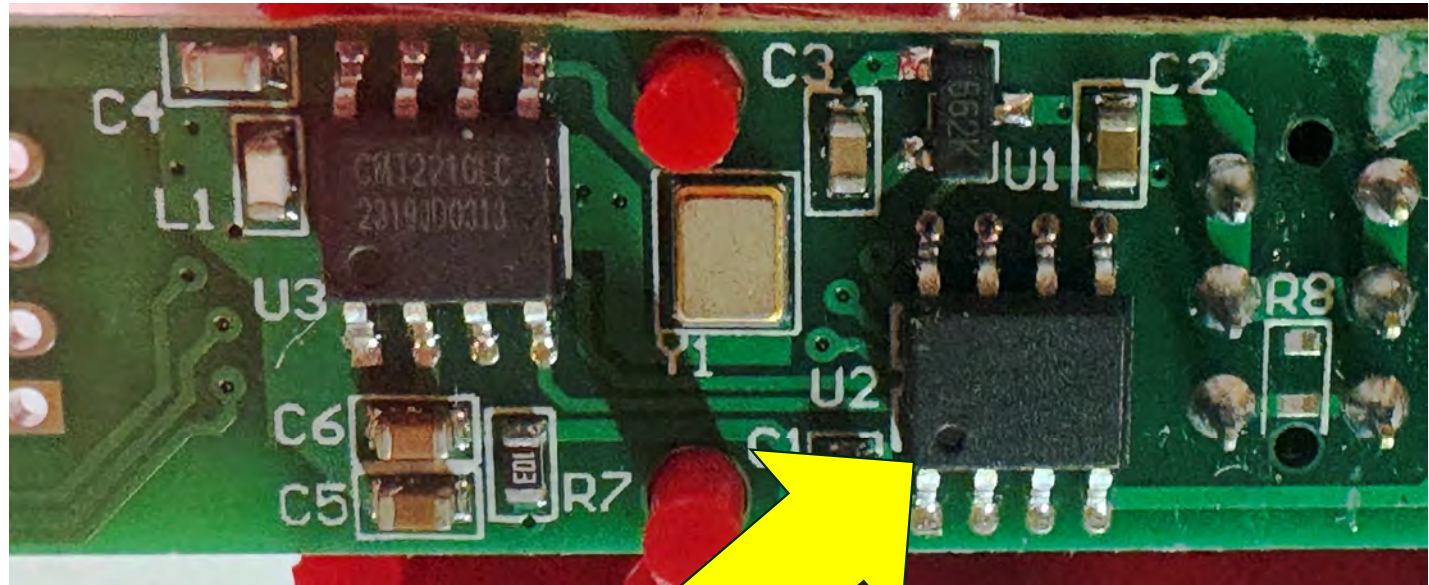
INTERNALS



Y1: 27.1383 MHz Crystal

Enables 433.92 MHz Reception Frequency

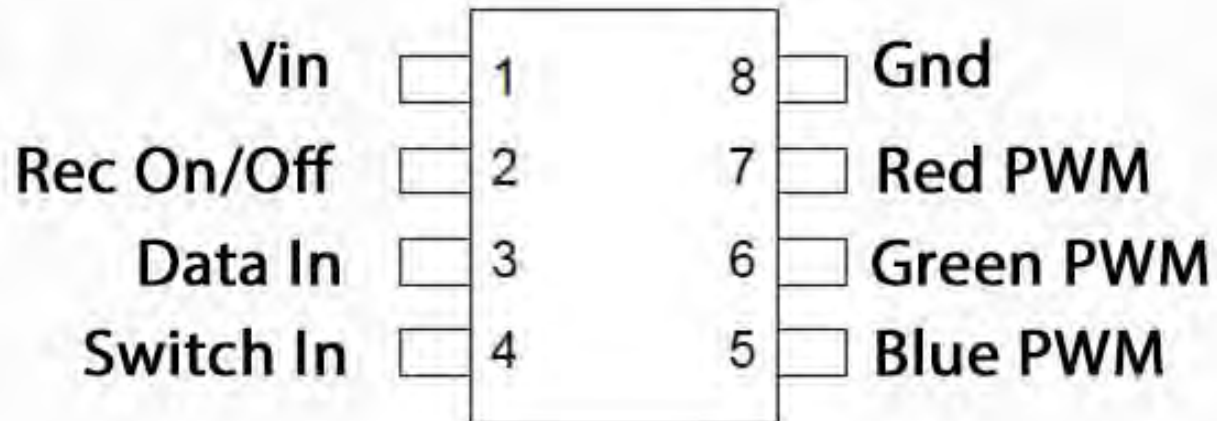
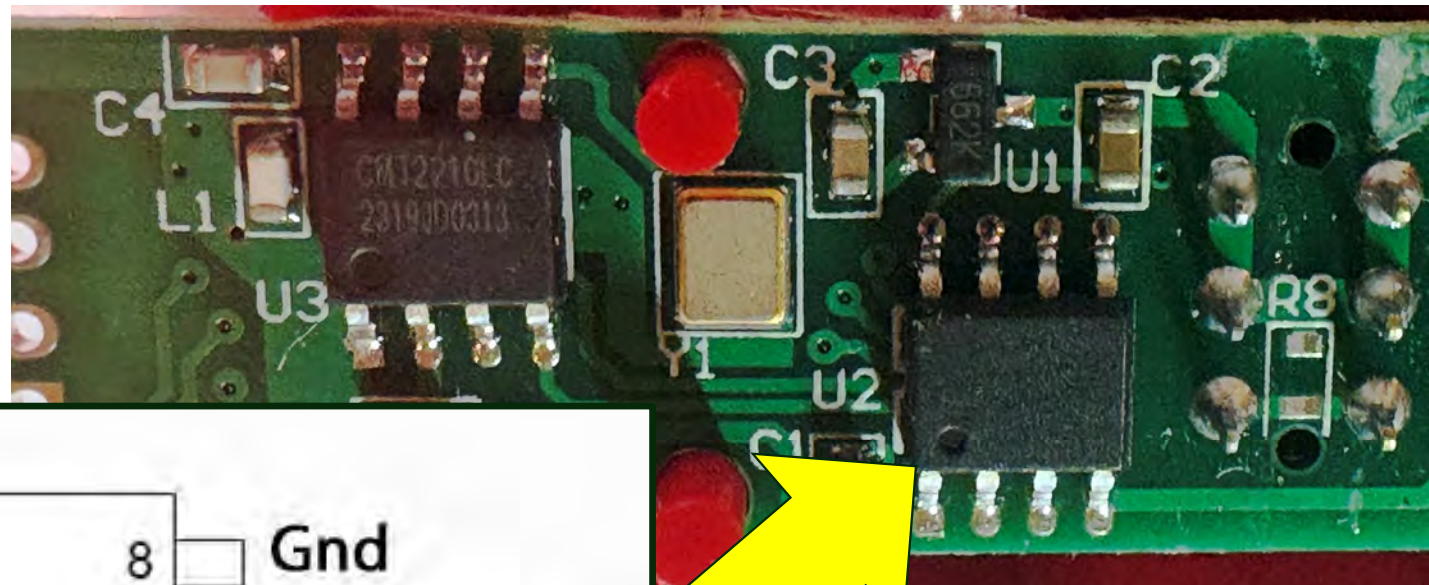
INTERNALS



U2: Unlabeled SOP-8 Package

Inexpensive OTP microcontroller

INTERNALS





REAL WORLD ISSUE

- Automotive Key Fob DOS
- Solution: intermittent transmissions

DMX OVER RF - SAMPLES

- HackRF One:
 - 433.9200 MHz at 50MHz bandwidth, 500ksps, c16 format
- Flipper Zero
 - Sub GHz -> Read Raw
 - 433.92 AM 650 R-int 0 spl
 - (Place in SD Card/subghz Directory)





FLIPPER ZERO REPLAY FILES

- Red
- Blue
- Green
- White
- Flash R G B
- Off
- Crossfade
- Rotate R B G



FLIPPER ZERO PLAYBACK

- Sub-Ghz Menu
- Saved
- [select stored file]
- Send





FLIPPER ZERO PLAYBACK



UNOFFICIAL CHALLENGES!

- 1) Document the 433.92 MHz RF signal. Describe the demodulation and control protocol
- First comprehensive post tagging me:
 @rgblights.bsky.social or
 @RGB_Lights

UNOFFICIAL CHALLENGES!

- 2) Create a Flipper Zero app to control light wands natively (no capture replay).
- First working App with Source code tagging me:



@rgblights.bsky.social or



@RGB_Lights

PRIZES FOR EACH CHALLENGE





CHRISTMAS LIGHT RESOURCES

- Xlights Software
- Xlights Support Group
- Falcon Controllers
- Genius Controllers
- Pixels and supplies #1
- Pixels and supplies #2
- LearnChristmasLighting.com



THANKS!

- Kate Kuhn & Worldwide Technologies
- Bob Ackerman & DataTribe
- Wasabi, Kiwi, Kyle Skey, David Maynor



RESOURCES:

<https://lightsatshmoo.free.nf/>

